

Lesson Plan: Kohiki Slip Decoration for Slab Building

Objectives

- Introduce participants to the history and aesthetics of kohiki slip decoration.
- Teach slab-building fundamentals with emphasis on surface treatment.
- Guide students through applying kohiki slip and integrating it into functional or sculptural forms.

Activities

1. **Introduction (20 min):** Brief history of kohiki slip; show examples.
2. **Slab Preparation (20 min):** Demonstrate rolling slabs evenly; discuss thickness control.
3. **Slip Application (60 min):** Demonstrate brushing, dipping, or pouring slip; encourage freedom and experimentation—see discussion and exercises below. Before lunch break, demonstrate stretching technique to distort brushwork for dramatic effect.
4. **Lunch and Patience (≈1.5 hr):** Allow time for slip and clay to firm up.
 - **Structured Lunch/Teapot Demo (40 min):** Provide entertainment in creating a teapot during lunch.
 - **Structured Lunch/Practice Slot (20-30 min):** Provide short guided exercises students can do while waiting.
 - Goal:** Turn downtime into deliberate practice focused on slip observation and simple brushwork.
5. **Slab Construction (1 hr):** Demonstrate joining slabs into simple forms; emphasize simple forms, scoring, and compression.
6. **Wrap-Up Discussion (30 min):** Share results. Encourage discussion and reflection on slip application and surface effects; consider surface effects from the firing process.

Teaching Notes

- Stress patience: kohiki slip rewards subtlety.
- Encourage integration of surface decoration with form.
- Highlight how firing atmosphere alters slip appearance.

Supplies, Recommended Tools, & Materials to Bring

Materials Supplied by Workshop Instructors, but in limited quantities.

- Clay body (dark stoneware recommended).
- White slip (prepared to a thick cream consistency).
- Rolling pins or slab roller.
- Brushes, sponges, etc.... †
- Straight edges, rulers, tar paper for patterns. †
- Canvas for rolling slabs. †
- Wooden boards for working surfaces. †
- Bevel cutters, knives, and scoring tools. †

Personal Tools Suggested for Participants to Bring to the Studio

- Apron or old clothes.
- Notebook and pen for notes and sketches.
- Personal brushes for slip application. Old crusty brushes are my favorite for creating interesting textures.
- Soft rib or plastic card for smoothing slabs.
- Any preferred carving or texturing tools.
- Any items marked with † above, if you have them.

Clothing Suggested for Participants at the Kiln Site

- Cotton or woolen clothing (avoid synthetics).
- Notebook and pen for notes and sketches.
- Sturdy, closed-toe shoes.
- Gloves for handling hot items and for handling rough materials.
- Hair ties, hats, or other means to keep long hair secured.

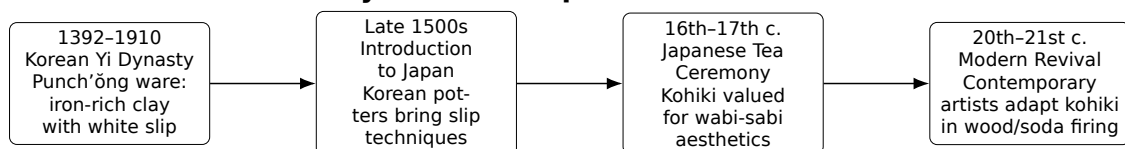
Here's a quote that I'd like everyone to keep in mind as we work through the kohiki slip exercises and slab building:

Without consciously thinking something is good or bad, creating as if it were the most natural thing in the world to do, making things that are plain and simple but marvelous, this is the state of mind in which artisans do their finest work.

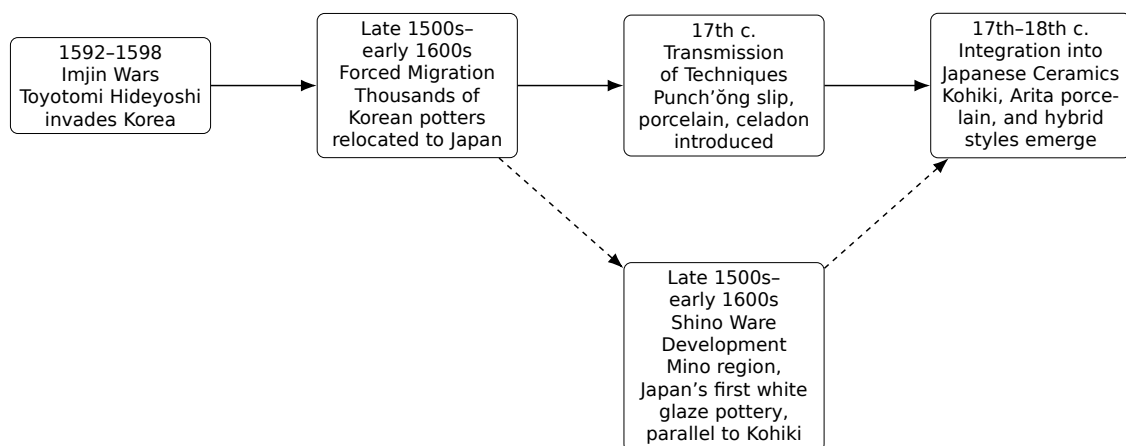
I think I can clarify this state of mind through an analogy. For human beings, walking is the most ordinary of activities. Even the most simple minded person can walk well. But no one is ever praised for being a good walker, and no one takes special pride in doing so. This is how I would like to view the artisan's state of mind when he is at work.¹

Kohiki Slip

Timeline: History of Kohiki Slip



Parallel Timeline: Import of Korean and Chinese Potters to Japan



¹Adapted from Soetsu Yanagi, **The Beauty of Everyday Things**, 1964.

A couple modern examples of popular kohiki slip decoration by Akira Satake and Robert Fornell, as shown in the images below. I started experimenting with kohiki slip decoration after seeing Satake's work.



A beautiful example of a more traditional kohiki piece is this tea bowl identified as Agano ware on this website: <https://qualitychanoyu.com/2023/01/13/antique-agano-ware-kobiki-chawan/>

ANTIQUÉ AGANO SHIRO-KOHIKI CHAWAN



Although Europeans later came to prize East Asian porcelain and re-shaped its markets, the white surfaces used in Punch'ōng and kohiki developed within East Asia before Europeans had sustained contact. These whiteslip techniques grew from local aesthetics and kiln practice, not from European influence. European demand in the 17th-19th centuries expanded export production in China and inspired European porcelain manufacture, but it did not influence Korea's (and later Japan's shino ware) whiteslip aesthetics.

Other links of interest about kohiki ware:

- <https://turuta.jp/ceramics-story/archives/21164>
- <https://dawan-chawan-chassabal.blogspot.com/2010/02/powdery-matsudaira-at-hatakeyama-museum.html>
- <https://dawan-chawan-chassabal.blogspot.com/2010/11/powdery-matsudaira-revisited.html>

After we work through the slabs, the following section will become apparent why we need to think about thickness of the slabs we are making.

Slab Thickness: Importance and Tradeoffs

The thickness of your clay slab is important for both the construction process and the final appearance of your work.

- **Structural Integrity:** Thicker slabs are less prone to warping and cracking during handling and drying, making them more forgiving for experimental work or for larger forms.
- **Workability:** Thinner slabs are easier to bend, shape, and join, but require more careful handling to avoid tearing or distortion during firing.
- **Surface Techniques:** Very thin slabs may dry too quickly for certain slip or texture applications, while thick slabs can retain moisture longer, allowing more time for surface work.
- **Aesthetics:** The visual weight and tactile feel of the finished piece are directly affected by slab thickness. Thicker walls can feel more robust, while thinner ones may appear more delicate or refined.

Tradeoff: Choose slab thickness based on your intended technique and form. For kohiki slip decoration, a medium thickness (typically 6-8 mm) balances workability, surface quality, and structural strength. Always test and adjust based on your clay body and project needs.

Exercises With the Brush

The following are exercises I encourage you to practice. We will work through them here, but please take these with you when you leave and work on them in your own studio time.

Practice (30–40 min)

Keep materials minimal: work on wooden bats for slip applications.

1. **Single-Stroke Studies (6 min)**: Make one confident stroke across the bat—experiment with pressure and angle.
2. **Opacity Ladder (6 min)**: Apply 3 horizontal bands with decreasing slip thickness (heavy → medium → thin). Let dry and note differences.
3. **Quick Pair Share (3 min)**: Swap bats with a partner; each gives one suggestion for the other's next step.
4. **Identify Area for Cutting and Construction (3 min)**: On your bat, mark an area you might cut out later for slab construction; consider how the slip application will interact with form.

Brushwork Patterns (Step-by-step)

1. **Stippling (6 min)**: Load brush, dab vertically using the tip only; vary density for tonal control.
2. **Drag-and-Stop (6 min)**: Pull brush in a steady stroke, lift quickly for crisp ends.
3. **Combing (2 min)**: Use a fork or comb dragged through semi-wet slip for parallel texture.
4. **Ribs (2 min)**: Drag a rib or flat tool through wet slip to create texture; vary angle and pressure for different effects.

Slip Application Tests

- **Brush-Load Test**: Dip brush and make a 10 cm stroke; note opacity and pooling.
- **Adhesion Test**: Apply slip to a scrap of the class body, bend gently when leatherhard—does it crack or flake?
- **Drying Rate**: Make a thin swatch and time until it is no longer tacky—record for next session.
- **Hypothesis**: Suggest one hypothesis on slip application and test.

Observation Prompts

- Changes in color/value as the slip dries.
- Areas of pooling vs even coverage.
- Edges where slip has feathered or pulled away.
- Any textural interaction where slip meets tool marks.

Lunch / Patience

Below are some more exercises I want you to try during lunch or downtime. These are designed to help you build your skills and understanding of kohiki slip decoration. We will also be talking a bit about various subjects during this time.

Exercise: Slab Forming and Stretching

For a more exaggerated & dramatic appearance of the brush work on your slab pieces, try the following exercise during lunch or downtime:

- **Slab forming:** Get a small slab of clay (about 8 mm thick) from the instructor with kohiki slip already applied.
 - **Stretching:** Carefully pick the small slab up from one edge and practice stretching it by carefully throwing it back down onto the table or canvas at roughly a 45 degree angle. The goal is to elongate the slab and distort the brushwork in an interesting way without tearing the clay.
 - **Repeat:** Try this several times, varying the side you pick up from and the angle of impact. Observe how the brushwork changes with each stretch.
 - **Notes:** Take a moment to write down your impressions about how to throw the clay to get different effects, and any challenges you faced in keeping the slab intact and well textured.
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Quick Recipe & Slip Consistency (for reference)

Basic Δ 10 Kohiki Slip Recipe	
EPK kaolin	500 g
Ball clay	400 g
Feldspar	600 g
Silica	500 g

Water: variable – depends on deflocculant^a.

1. Toothpaste consistency.
2. Thick cream consistency.
3. Any range in between depending on your application style and decorative goals.

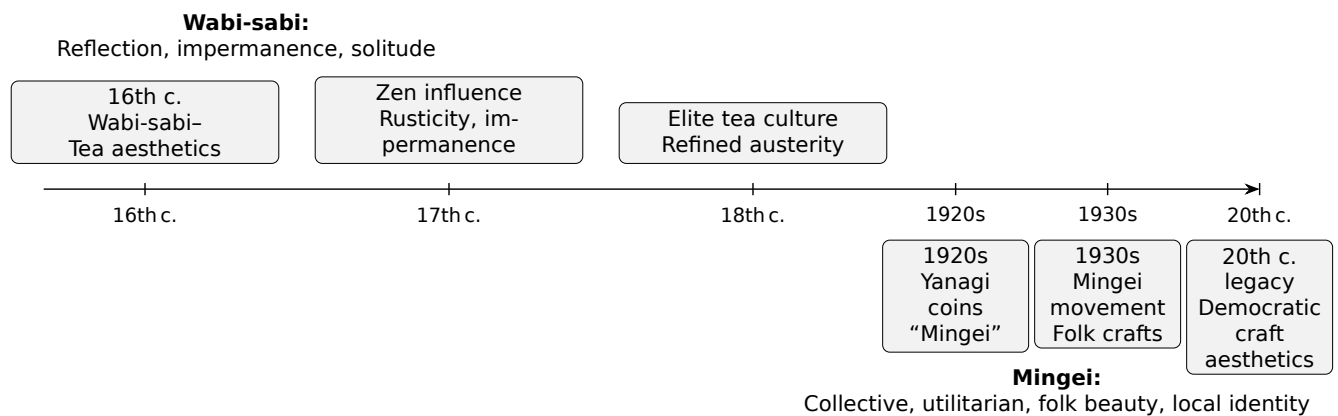
^aDeflocculant (sodium silicate or Darvan): 1–3 g if needed, to thin without settling. Deflocculants help reduce the amount of water needed, but can reduce the workability of the clay beneath the slip. Use sparingly and test.

Lunch Discussions

Aesthetics of Kohiki

Discuss wabi-sabi, the value of subtlety, and how controlled imperfection enhances form.

Wabi-sabi and Mingei



Aspect	Mingei (Yanagi)	Wabi-sabi
Origin	1920s-30s folk craft movement led by Yanagi Sōetsu	16th century Zen tea aesthetics
Focus	Everyday, utilitarian objects made by an anonymous craftsman*	Imperfection, impermanence, rustic simplicity
Philosophy	Collective recognition of "beyond beauty and ugliness"	Contemplative, insight into transience
Social Dimension	Communal craft traditions, cultural preservation	Elite tea culture, spiritual refinement

*Please understand I am using the term "craftsman" in the same form as "human."

How Yanagi Differentiated Them

- **Mingei as collective beauty:** Yanagi argued that folk crafts are beautiful precisely because they are made by nameless artisans for everyday use, not for individual artistic expression. This contrasts with *wabi-sabi*, which often highlights the personal, contemplative experience of imperfection.
- **Functional vs. contemplative:** Mingei objects are functional tools of daily life (bowls, textiles, furniture). Wabi-sabi objects (tea bowls, gardens, architecture) are often vehicles for meditation and aesthetic reflection.
- **Democratic vs. elite aesthetics:** Yanagi saw mingei as a community art form, accessible to all. Wabi-sabi, though humble, was historically tied to the tea ceremony and elite Zen culture.

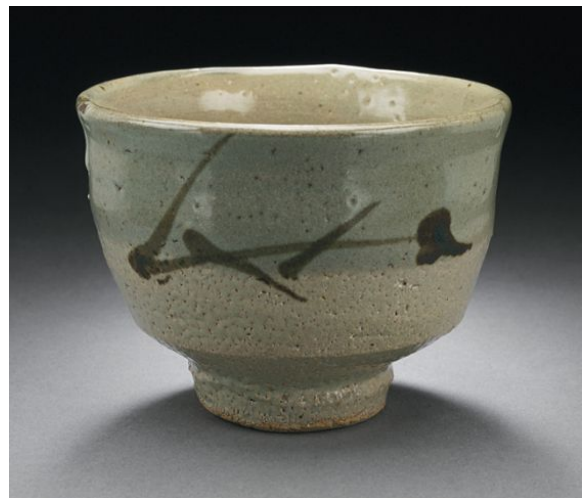
Why This Matters

Yanagi's distinction is important because it shows that there are different ways of thinking about beauty—some focus on everyday objects made by many people (mingei), while others focus on quiet, simple beauty, and personal reflection (wabi-sabi). Mingei is about community, utility, and cultural survival, while wabi-sabi is about solitude, impermanence, and spiritual refinement. Both overlap in valuing simplicity and naturalness, but they diverge in social function and philosophical grounding.

Put simply, Mingei is a social philosophy of creation and use, while Wabi-sabi is *a way of seeing the world* and objects within it. Objects do not possess a quality of wabi-sabi; rather, wabi-sabi is a lens through which we appreciate them.



Images exemplifying the wabi-sabi aesthetic.



Selected works by Hamada and Kawai, exemplary Mingei works.

Kiln Transformations

The basic principles of firing a kiln apply across all types of kilns. The most fundamental component across all kiln types for creating a ceramic piece involves transforming raw clay into a durable, vitrified material through controlled heating. This is true for our process where we bisque fire the pieces first and again when we glaze fire them. In an electric kiln, the heat is supplied via electric current through heating elements similar to any modern electric oven, electric stovetop, or electric toaster. In a gas kiln, the heat is generated by burning a fuel source such as natural gas or propane. In both electric and gas firings, we have the atmosphere inside the kiln; with electric, excess oxygen exists in the atmosphere of the kiln since combustion is not the local heating source,² while in gas firings, we can adjust the air-to-fuel ratio to allow excess oxygen (oxidation) or limit oxygen (reduction or neutral) within the kiln atmosphere.³

As opposed to firing with an electric or gas kiln, wood and soda firings introduce unique atmospheric conditions that significantly affect the final appearance of ceramic pieces. In the modern age, electric kilns and gas kilns lack the dynamic interactions present in wood and soda firings.

Wood Firing Basics

Wood firing refers to the fuel source being wood. As the wood combusts, it releases heat (needed for the transformation of the bisque ceramic and glaze vitrification), H_2O , CO_2 (among other carbon byproducts), and ash particles that circulate within the kiln. These ash particles from wood as the fuel source are what we refer to as “the atmosphere” and they settle on the surfaces of the ceramic pieces during the firing process.

When the temperature of the atmosphere in the kiln and the surface of the ceramic pieces climb above 1150°C (2100°F) the ash reacts with the silica, alkali and alkaline earth elements, and alumina on the surface of the ceramic piece, forming a natural glaze. The longer the firing continues at high temperatures, the more ash is deposited, leading to thicker and more varied glaze effects. The placement of pieces in the kiln also affects how much ash they receive, with those closer to the firebox typically getting more coverage.

Soda Firing Basics

When soda firing, we introduce a soda solution (commonly a mixture of sodium carbonate and sodium bicarbonate dissolved in water, and buffered with calcium carbonate) into the kiln at high temperatures, typically around

²Unless your electric energy source is from solar, geothermal, wind, or hydroelectric power, the electric kiln is still indirectly powered by combustion of fossil fuels, i.e. diesel generators.

³For reduction, excess fuel is being combusted producing carbon products lacking in oxygen, but containing an extra electron, which readily exchanges with chemically bound oxygen in the clay bodies (body reduction) and oxygen in the glazes (glaze reduction).

900–1100°C (1650–2000°F). The soda decomposes upon contact with the hot kiln atmosphere, releasing sodium oxide gas (Na_2O) and carbon dioxide (CO_2). The sodium oxide gas reacts with the silica and alumina in the clay body and glaze, forming a complex sodium-alumino-silicate glaze on the surface of the ceramic pieces. This sodium oxide in the kiln is what referred to as “the atmosphere” associated with soda firing.

The important point to note here is that the soda can be introduced at any point during the firing process once the kiln reaches above the full decomposition temperature of the soda compounds (around 900°C). This means that the soda atmosphere can be introduced during the early stages of the firing or near peak temperature.⁴ Each timing choice will yield different surface effects on the ceramic pieces.

Wood + Soda

The best of both worlds! By combining wood firing and soda firing, we can achieve unique surface effects that leverage the benefits of both techniques. The wood ash contributes to the natural glaze formation, while the soda enhances the interaction of the ash with the clay body and glaze. The soda atmosphere can also help to create a more uniform glaze layer by reacting with the ash deposits, resulting in a smoother and more cohesive surface finish.

For the Spring Valley cross draft kiln, we will use a soda mixture of approximately 50% sodium carbonate, 30% sodium bicarbonate, and 20% calcium carbonate dissolved in water. This mixture will be introduced into the kiln at the late stages of the firing process by soaking pieces of wood and bark in the mixture and then placing this wood and soda mixture into the firebox. As the wood burns, it will release the soda into the kiln atmosphere, allowing it to interact with the ash and the surfaces of the ceramic pieces. We will also enforce greater reduction in the kiln by slowing the air intake to the firebox, which will limit the oxygen available for combustion.

Slab Expansion & Construction

Slab expansion is not required for kohiki slip decoration, but it can enhance the visual impact of the brushwork by distorting and stretching the applied slip. The process involves carefully picking up a slab of clay with kohiki slip already applied and throwing it back down onto a flat surface at an angle. This action can take some practice as there are many failures which can occur with this type of work.

In contrast to my work, Satake uses slabs which are quite thick and soft when he stretches them through a ‘dropping’ process. If throwing slabs at an angle is not successful for you, consider experimenting with softer slabs (more difficult to handle without tearing) or thicker slabs (more difficult to stretch). The goal is to find a balance that allows you to achieve interesting distortions of the brushwork without damaging the slab.

⁴Soda firing is accomplished at Δ6 firings quite successfully in outdoor kilns.

Choosing your Cuts

When selecting areas of your slab to cut out for construction, consider how the brushwork will interact with the form you plan to create. Look for areas where the slip application has interesting textures or patterns that will enhance the overall aesthetic of the piece. Additionally, think about how the cut pieces will fit together and how the brushwork will flow across the joins.

Joining Slabs

Some basic tips for consideration when joining slabs:

- Score both surfaces to be joined.
- I recommend a thin light layer of water in the scoring grooves to help with adhesion rather than using slip. I've found that slip will often extrude from the joint and create a messy appearance.
- Apply pressure along the joint to ensure a strong bond. I've found that compression is the key to a strong joint.
- Allow the joined piece to dry evenly to prevent warping or cracking.

Handles Handles get their own treatment here because I like to have texture on both sides of the handle. To achieve this, I will fold a thinly stretched piece of the slab over itself to create a good-feeling thickness for the handle. This way, both sides of the handle have kohiki slip decoration. After forming the handle, I score and compress the ends to the main body of the piece as described above.

Wrap-up; Reflection, Critique, Expectations, and Next Steps

I encourage you to think about the pieces you have made today and how you might approach similar work in the future.

- How do you think the wood and soda firing process will affect the appearance of your pieces?
- What aspects of the kohiki slip application did you find most successful?
- Were there any challenges you faced during the slab construction process?
- What would you do differently next time to enhance your work?
- Looking at the work completed here, what surface excites you the most?
- Looking at the work completed here, what are you most excited to see after the firing?
- What expectations do you have for how the kohiki slip will appear after firing?

Safety & Handling Notes

- Wash hands after handling slips containing deflocculants (the ones in the workshop do not have deflocculants!).
- Keep sanding and dry-scraping to a minimum.
- Label slip containers; avoid cross-contamination between different slips.

What I Notice / What I Would Change

1. Identification: _____

2. Observations: _____

3. One change I would make next time: _____

Recommended Reading Materials

- **Soda, Clay, and Fire** by Gail Nichols – Comprehensive guide on soda firing techniques and chemistry.⁵
- **The Beauty Everyday Things** by Yanagi Sōetsu – Foundational text on the mingei philosophy.
- **Wabi-Sabi for Artists, Designers, Poets & Philosophers** by Leonard Koren – Exploration of wabi-sabi aesthetics.
- **Making Marks: Discovering the Ceramic Surface** by Robin Hopper – Insights into surface decoration techniques.
- **Soda Glazing** by Ruthanne Tudball – Practical guide on soda glazing methods.
- **Salt Glazing** by Phil Rogers – While focused on salt glazing, it provides useful context for soda firing.
- <https://www.beckpots.com/process> – Casey Beck's website; Worth exploring for soda firing recipes & techniques.

⁵Out of print.

Lesson Plan: Wood-Fired Soda Firing

Objectives

- Review fundamental principles of firing across all types of kilns.
- Discuss the unique characteristics of wood firing.
- Explain the chemistry and effects of soda firing.
- Introduce participants to the principles of soda firing in a wood kiln.
- Teach preparation of work for soda atmosphere (wadding, glazing, placement).
- Explore how soda interacts with clay and slip surfaces.

Materials

- Bisque-fired student work (preferably with surfaces prepped for wood+soda).
- Wadding materials (refractory wadding mix, alumina hydrate, kiln wash).
- Kiln shelves, posts, protective gear.
- Wood for firing (species discussion: pine, oak, etc.).
- Soda mix (soda ash + baking soda + whiting, in water) – following the suggestions of Gail Nichols.

Activities

1. **Introduction:** Explain soda firing chemistry; show examples.
2. **Preparation of Work:** Demonstrate wadding placement; discuss glaze choices.
3. **Kiln Loading:** Explain loading strategies; emphasize safety and teamwork.
4. **Firing Process:** Walk through firing stages; demonstrate soda introduction; discuss temperature targets.
5. **Cooling & Unloading:** Explain cooling protocols; plan for unloading and critique.
6. **Wrap-Up Discussion:** Anticipate results; encourage embracing variation and unpredictability.

Teaching Notes

- Stress safety: protective gear, teamwork, kiln etiquette.
- Highlight unpredictability as part of the aesthetic.
- Connect back to constructed works: how soda atmosphere transforms surfaces.